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INTERIM PROJECT REPORT-I

on

ANVESH: Cyber Forensic Intelligence & Email Threat Detection Platform

(Aligned with Smart India Hackathon 2026 Problem Statement SIH26106)

Submitted in partial fulfillment for the award of

BACHELOR OF TECHNOLOGY

Degree in Computer Science & Engineering

Academic Session: 2026–27

Under the Guidance of:
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Date of Submission: October 2026

APPENDIX-1
PROJECT PROGRESS APPROVAL FORM (INTERIM REPORT-I / MTE)

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Academic Session: 2026–2027 **Semester:** 7th Semester (MTE)

Project Title: ANVESH: Cyber Forensic Intelligence & Email Threat Detection Platform

Problem Statement ID: SIH26106 (Smart India Hackathon 2026)

Student Team Candidate Information

S.No.	Student Name	Admission No.	Roll No.	Signature
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Project Progress Status at MTE Milestone

S.No.	Milestone Deliverable / Criterion	Status	Guide Remarks
1.	Problem Formulation & Definition	**COMPLETED**	Approved by Guide
2.	Literature Review & State-of-the-Art Survey	**COMPLETED**	Rigorous survey across IEEE/ACM
3.	Drafting Introduction Section of Research Paper	**COMPLETED**	Formulated in Chapter 1
4.	Drafting Literature Review Section of Research Paper	**COMPLETED**	Formulated in Chapter 2
5.	Identification of Specific Research Gaps	**COMPLETED**	5 distinct gaps defined
6.	Prototype Feasibility & Core Pipeline Verification	**IN PROGRESS**	RFC-822, ML & Npcap verified

Guide's Assessment & Recommendation:

This is to certify that the above students have satisfactorily carried out their Capstone Project Work entitled **'ANVESH: Cyber Forensic Intelligence & Email Threat Detection Platform'** during the 7th Semester up to the Mid-Term Examination (MTE) milestone. The progress of the project, including the Literature Review and Research Paper Introduction/Survey sections, has been evaluated and approved for submission.

[INSERT SIGNED PROJECT PROGRESS APPROVAL FORM FROM GUIDE — APPENDIX-1]

Signature of Guide: _____ **Date:** October ____, 2026

Name of Guide: [GUIDE NAME] **Designation:** [DESIGNATION]

SYNOPSIS

1. Project Background

Electronic mail communication serves as the fundamental operational backbone for modern enterprise collaboration, inter-organizational financial transactions, and official administrative governance. However, the foundational underlying protocols —namely SMTP (RFC 5321) and the RFC 5322 message format standard—were established during the nascent development of the internet with an emphasis on open interoperability over defensive security. Consequently, internet mail inherently lacks mandatory cryptographic sender identity authentication, non-repudiation, and transit integrity guarantees. Cyber adversaries exploit these structural vulnerabilities to deliver multi-stage social engineering campaigns, credential harvesting vectors, and fraudulent financial redirection lures. The Smart India Hackathon (SIH 2026) Problem Statement **SIH26106** specifically targets this challenge, demanding an integrated, verifiable cyber-forensic intelligence platform to perform automated email header analysis, threat deconstruction, and court-admissible evidence reporting.

2. Cybersecurity Problem

Conventional enterprise defenses remain fragmented across disconnected functional silos. Traditional Secure Email Gateways (SEGs) rely predominantly on static blacklists of known malicious domain names, IP addresses, and file signatures. When sophisticated attackers deploy Business Email Compromise (BEC) schemes—such as compromising a legitimate corporate account or registering an orthographically similar lookalike domain—traditional defenses suffer catastrophic failure. In an 'Authenticated BEC' attack, the sender transmits messages from a compromised authentic corporate mailbox, resulting in complete cryptographic passes across SPF, DKIM, and DMARC. Because conventional gateways equate cryptographic passes with benign intent, financial wire diversion lures bypass perimeter inspection entirely. Furthermore, Security Operations Center (SOC) tooling completely bifurcates Layer 7 email threat analysis from Layer 3/4 network transport telemetry, failing to correlate post-delivery Command-and-Control (C2) beaconing or unauthorized connection anomalies with inbound email lures. Finally, existing machine learning detectors function as unexplainable 'black boxes,' generating opaque probability scores while failing to preserve an unbroken, cryptographically verifiable chain of custody necessary for legal prosecution.

3. Motivation

The motivation behind ANVESH (■■■■■■) arises from the severe operational friction experienced by SOC analysts and law enforcement investigators. Investigators must manually pivot across multiple disparate command-line MIME utilities, external DNS/WHOIS portals, commercial threat intelligence APIs, Wireshark packet analyzers, and manual text processors. This fragmented workflow creates operational latency, often delaying incident containment by hours or days while increasing analyst alert fatigue. More critically, informal evidence extraction breaks evidentiary continuity, resulting in judicial dismissals during legal proceedings. ANVESH is motivated by the necessity for a unified, zero-fabrication cyber forensic workbench that synthesizes multi-tier threat detection, network transport telemetry, transparent explainable risk scoring, and automated court-admissible forensic reporting.

4. Proposed ANVESH Solution

ANVESH implements a modular, defense-in-depth architecture. The platform ingests raw RFC-822 (.eml) payloads and immediately establishes an immutable SHA-256 evidence baseline. It parses multipart MIME boundaries, traverses chronological Received: hops backwards to candidate origin gateways, and validates cryptographic SPF, DKIM, and DMARC parameters alongside BGP autonomous system numbers (ASN) and Tor exit node registries. To avoid the pitfalls of a monolithic classifier, ANVESH deploys a specialized multi-tier AI/ML subsystem comprising: Model 1 (TF-IDF + Calibrated Logistic Regression for phishing credential lures), Model 2 (Balanced Logistic Regression for BEC financial coercion and wire fraud keywords), Model 3A (Deterministic heuristic discrepancy matching for display-name spoofing), Model 3B (Random Forest structural predictor on Levenshtein and Jaro-Winkler distances against high-value target brand registries), and Model 4/5 (Random Forest classifier trained on 41 NSL-KDD transport features hooked to passive Npcap live wire packet sniffing). All outputs are synthesized through an Explainable Signal Fusion Engine that yields a normalized 0–100 threat score based on transparent, auditable point contributions.

5. Major Objectives

The primary objectives of the project are: (1) To architect an automated RFC-822 ingestion parser that unrolls MIME structures and commits artifacts to an immutable SHA-256 evidence ledger; (2) To implement reverse chronological Received: hop traversal to reconstruct message transit routing across autonomous systems; (3) To evaluate cryptographic SPF, DKIM, and DMARC authentication records to differentiate administrative misconfigurations from active spoofing; (4) To train specialized machine learning models for lexical phishing detection and BEC financial urgency analysis; (5) To construct a lookalike domain detector evaluating Levenshtein and homoglyph metrics; (6) To integrate low-level network session telemetry (Engine 5) using the 41-feature NSL-KDD benchmark; (7) To implement an Explainable Signal Fusion engine enforcing the Attribution Boundary Safeguard ('Actor Identity: NOT ESTABLISHED'); and (8) To automate the generation of court-admissible forensic PDF dossiers containing complete evidence digests.

6. High-Level Methodology

The project methodology follows an empirical engineering lifecycle: (1) Ingestion & Evidence Hashing: Ingesting raw RFC-822 files and generating SHA-256 hashes committed to an append-only audit trail; (2) Header Parsing & Routing Reconstruction: Decoding MIME headers, traversing Received: hops, resolving BGP ASNs, and evaluating cryptographic authentication matrices; (3) Multi-Model Machine Learning Inference: Running specialized models in parallel for phishing text, BEC urgency, and lookalike domain detection; (4) Network Telemetry Sniffing: Passive packet sniffing via Npcap, 5-tuple flow aggregation, and classification through Model 4/5; (5) Bayesian Signal Fusion: Synthesizing multi-vector scores into an explainable 0–100 risk score while enforcing evidentiary neutrality; and (6) Report Generation: Automated compilation of cryptographically sealed forensic dossiers using ReportLab.

7. Current Development Status (Interim MTE Milestone)

At the 7th Semester MTE milestone: The comprehensive literature review across email phishing, BEC, lookalike domains, network intrusion benchmarks, and digital evidence standards has been fully completed. The foundational Introduction and Literature Review sections of the prospective research paper have been drafted. Five foundational research gaps have been identified and defined. The core functional modules—including RFC-822 parsing, SHA-256 hashing, Models 1, 2, 3A, 3B, 4/5 (NSL-KDD Random Forest), Scapy Npcap live wire sniffing, Signal Fusion scoring, and ReportLab PDF report generation—have been implemented and verified within the repository. The React 18 + Vite dark-mode SOC dashboard is operational and communicates with the FastAPI backend.

8. Expected Contribution

Upon final completion, ANVESH will deliver: (1) A unified, zero-fabrication forensic intelligence platform eliminating tool fragmentation for SOC analysts; (2) An effective defense against 'Authenticated BEC' attacks by decoupling behavioral threat modeling from cryptographic transport authentication; and (3) A legally sound digital forensic framework that strictly separates technical infrastructure observables from human actor attribution, guaranteeing legal admissibility.

9. Current Limitations

The current interim prototype possesses acknowledged limitations: (1) Language Specificity: Models 1 and 2 are trained on English-language corpora; (2) Passive Sniffing Throughput: Python-based Scapy packet capture is suitable for desktop and local testbed traffic but lacks DPDK kernel-bypass acceleration for multi-gigabit enterprise trunks; (3) Threat Intel Connectivity: Deep reputation enrichment relies on external network access, falling back to local heuristics in air-gapped environments; and (4) Non-Attribution Ceiling: The system strictly maps technical infrastructure and transit routing, never physical human operators without lawful subscriber subpoenas.

10. Future Direction

Subsequent project phases leading to the End-Term milestone will focus on: (1) Implementing optical extraction for QR-code phishing ('Quishing') lures via computer vision; (2) Building an automated Section 65B Electronic Evidence Certificate generator compliant with Indian Evidence Law; (3) Conducting formal high-volume benchmark testing; and (4) Finalizing the full Capstone Project Report and submitting the empirical research paper.

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CHAPTER 1
INTRODUCTION

1.1 Background of the Problem

Electronic mail remains the foundational operational infrastructure for modern digital commerce, administrative governance, and interpersonal enterprise collaboration. Over four billion users generate in excess of three hundred billion email transmissions daily, forming an indispensable web of operational coordination. However, the foundational architectural protocols governing electronic mail—specifically the Simple Mail Transfer Protocol (SMTP) formalized in RFC 821/5321 and the standard message representation defined in RFC 822/5322—were engineered during the nascent development era of the ARPANET. The paramount design imperative of that era was friction-free, distributed interoperability among mutually trusted institutional research nodes. Consequently, fundamental internet mail protocols were established without intrinsic security primitives, lacking mandatory cryptographic sender identity authentication, content confidentiality, and transmission non-repudiation.

This structural absence of native trust verification has established electronic mail as the predominant vector for advanced cyber attacks globally. According to the Federal Bureau of Investigation (FBI) Internet Crime Complaint Center (IC3) and the Verizon Data Breach Investigations Report, over 80% of reported enterprise security breaches originate through social engineering schemes delivered via email [1], [2]. The contemporary threat landscape has evolved far beyond rudimentary mass-spam campaigns characterized by obvious grammatical anomalies and generic greeting structures. Modern adversaries deploy multi-stage, highly targeted intrusion campaigns categorized into sophisticated threat vectors:

- **1. Advanced Phishing & Credential Harvesting:** Attackers leverage highly context-aware social engineering lures tailored to specific organizational workflows. Obfuscated hyperlinks direct victims to high-fidelity replica portals engineered to capture credentials and multi-factor authentication tokens.
- **2. Business Email Compromise (BEC) & Financial Diversion:** Rather than relying on malicious file attachments or suspicious external URLs, BEC schemes employ targeted, text-only communications mimicking corporate executives, legal counsel, or external vendors. Adversaries exploit psychological pressure, urgency, and operational hierarchy to coerce accounting personnel into executing unauthorized wire transfers or redirecting vendor payment accounts [3].
- **3. Display-Name & Identity Spoofing:** Because RFC 5322 explicitly separates the user-friendly 'Display Name' from the technical envelope 'Mailbox Address,' attackers routinely configure fraudulent messages where the display name reflects a trusted executive identity while the mailbox belongs to an unrelated external address.
- **4. Homoglyph & Typosquatted Lookalike Domains:** Threat actors register domains that visually mimic legitimate enterprise brands by exploiting orthographic character substitutions, hyphen insertions, or Internationalized Domain Name (IDN) homoglyphs (e.g., substituting the Latin character 'l' with the digit '1').
- **5. Multi-Hop Transport Obfuscation & Relay Abuse:** Advanced threat campaigns route inbound messages through complex chains of open proxies, compromised mail servers, bulletproof hosting providers, and Tor exit nodes, systematically obfuscating the authentic

geographical and network origin of the payload.

- **6. Transport-Layer Network Exploitation:** Cyber threat campaigns rarely terminate at the inbox. In multi-stage attacks, social engineering lures serve as an initial delivery mechanism for malware that establishes persistent Command and Control (C2) communication channels, executes lateral network reconnaissance, or initiates Denial-of-Service (DoS) floods across the enterprise perimeter.

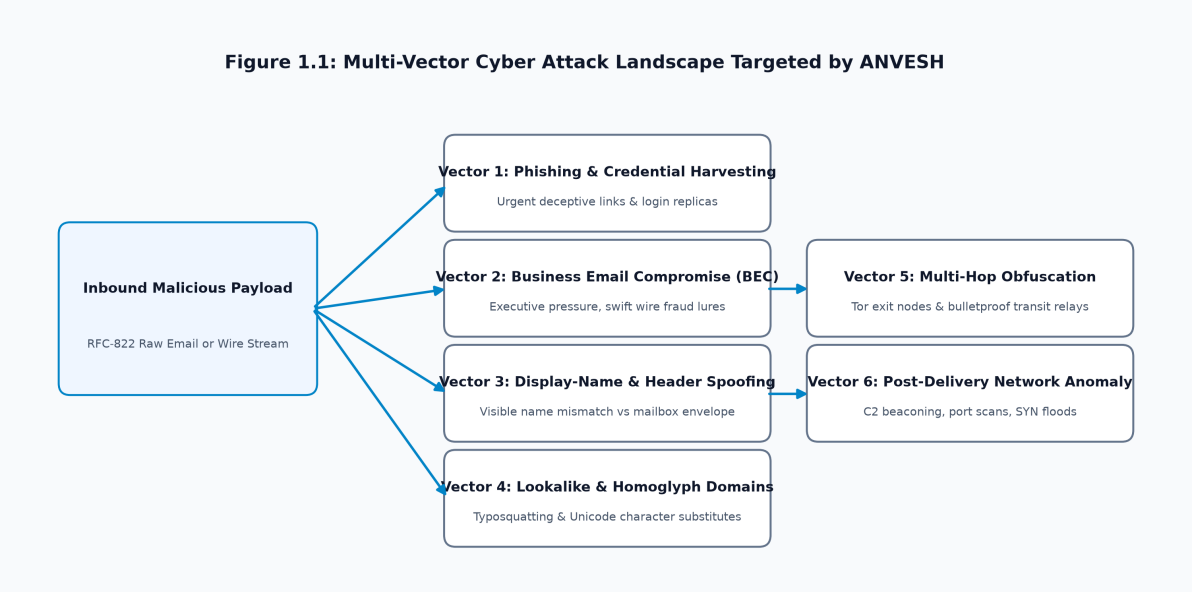


Figure 1.1 Conceptual Multi-Vector Cyber Attack Landscape: Phishing, BEC, Homoglyphs, and Network Anomalies

1.2 Motivation and Significance

The motivation behind ANVESH (■■■■■■) arises from a fundamental operational vulnerability in modern cybersecurity defense: the fragmentation of threat detection and cyber-forensic investigation tools. In contemporary Security Operations Centers (SOCs) and Law Enforcement Cyber Crime Units, analysts are confronted with disconnected software ecosystems. Application-layer email inspection is managed by cloud email gateways; domain reputation is checked through third-party WHOIS and threat intelligence portals; network anomaly detection is conducted via isolated intrusion detection systems; and evidence preservation is performed using manual hashing scripts. When an incident occurs, investigators must manually pivot across multiple disjoint consoles: unrolling headers in text editors, resolving autonomous system numbers (ASNs), submitting indicators to public databases, inspecting Wireshark captures, and drafting forensic summaries in manual word processors.

This fragmented workflow introduces severe operational latency and acute evidentiary risks: (1) **Fragile Chain of Custody:** Manual copying and screenshotting breaks cryptographic continuity, leading to dismissals during legal court cross-examinations; (2) **The 'Authenticated BEC' Blindspot:** Commercial gateways frequently treat cryptographic SPF/DKIM/DMARC passes as definitive proof of benign intent, failing to intercept wire fraud sent from compromised genuine corporate accounts; (3) **Opaque Black-Box Classifiers:** Deep neural networks output uncalibrated probability scores that cannot be explained or defended in a court of law; and (4) **Premature Attribution Fallacies:** Tools frequently accuse the owner of an IP address found in transport headers, falsely blaming innocent intermediate server administrators. ANVESH is motivated by the

urgent necessity for a unified, zero-fabrication cyber forensic intelligence workbench that resolves these critical gaps.

1.3 Problem Statement

The formal engineering problem statement of the ANVESH project is formulated as follows: *How to design, implement, and validate an integrated cyber-forensic intelligence platform that systematically ingests, parses, and evaluates suspicious RFC-822 email payloads and associated network transport telemetry; detects multi-vector threats (phishing, business email compromise, identity impersonation, lookalike domains, and network session anomalies) using explainable, multi-tier machine learning; preserves strict cryptographic evidence integrity via an immutable SHA-256 audit ledger; and produces court-admissible forensic dossiers while enforcing an ethical invariant that separates transport infrastructure observables from physical human actor attribution?*

A foundational principle governing the platform is the **Attribution Boundary Safeguard**: Technical transport headers, routing hops, BGP prefixes, and IP address resolutions identify transit infrastructure and network gateways only. Physical human identity cannot be established solely from transport-layer observables without lawful ISP subscriber log subpoenas. ANVESH enforces this evidentiary principle across all analytical modules, explicitly declaring: '**Actor Identity: NOT ESTABLISHED**'.

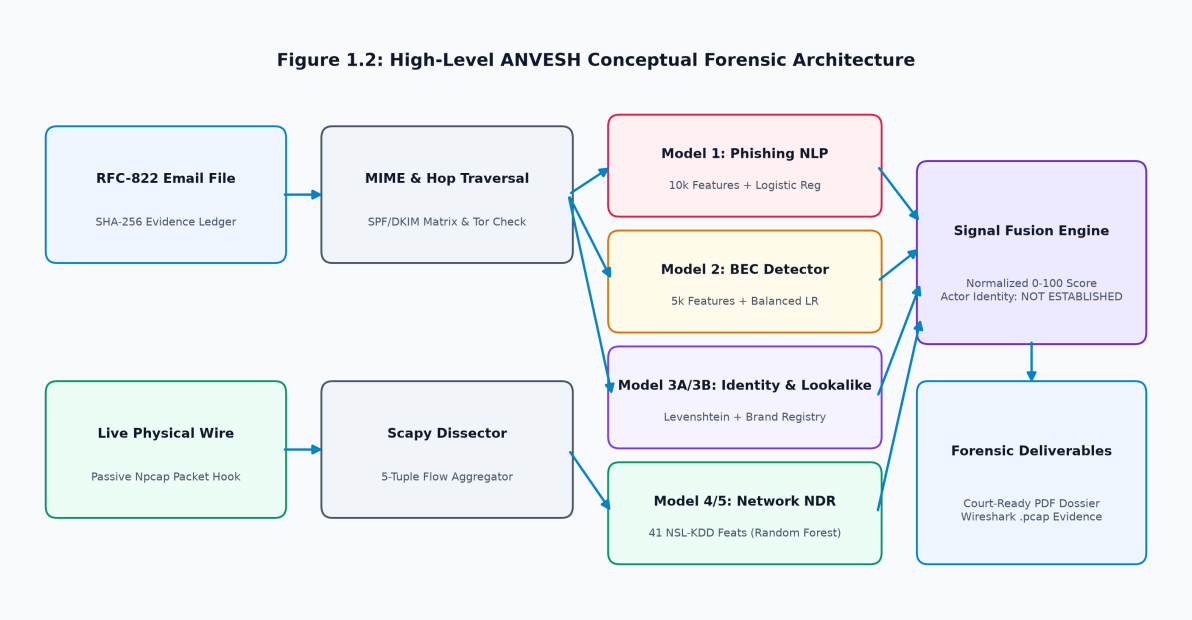


Figure 1.2 High-Level Conceptual Overview of the ANVESH Unified Forensic Intelligence Architecture

1.4 Objectives of the Project

The technical and academic objectives of the ANVESH platform, directly mapped to implementation modules, are defined in Table 1.1:

Table 1.1 ANVESH Project Objectives and Technical Implementation Alignment

Obj ID	Detailed Project Objective	Mapped Subsystem / Module	Status
OBJ-01	Automated RFC-822 ingestion parser with SHA-256 evidence digests.	backend/app/endpoints/emails.py	**VERIFIED**
OBJ-02	Reverse Received: hop traversal to reconstruct message routing.	backend/app/services/attribution.py	**VERIFIED**
OBJ-03	Cryptographic matrix evaluating SPF, DKIM, and DMARC records.	backend/app/services/forensic_fusion.py	**VERIFIED**
OBJ-04	Train NLP vector model (Model 1) detecting credential lures.	ml/models/phishing_baseline_v1/	**VERIFIED**
OBJ-05	Train specialized BEC detector (Model 2) for financial fraud.	ml/models/bec_baseline_v1/	**VERIFIED**
OBJ-06	Develop identity discrepancy engine (Model 3A) for display-name spoofing.	backend/app/services/identity.py	**VERIFIED**
OBJ-07	Build lookalike domain classifier (Model 3B) via Levenshtein metrics.	ml/models/lookalike_domain_v1/	**VERIFIED**
OBJ-08	Train network intrusion model (Model 4/5) on 41 NSL-KDD features.	ml/models/network_intrusion_v1/	**VERIFIED**
OBJ-09	Passive packet sniffing via Npcap driver for live wire capture & PCAP.	backend/app/services/scapy_service.py	**VERIFIED**
OBJ-10	Explainable Signal Fusion engine synthesizing multi-model scores (0-100).	backend/app/services/forensic_fusion.py	**VERIFIED**
OBJ-11	Automate cryptographically sealed court-admissible PDF dossiers.	backend/app/services/pdf_report.py	**VERIFIED**
OBJ-12	Build responsive web SOC workstation with visual triage and cases.	web/src/pages/DashboardPage.tsx	**VERIFIED**

1.5 Scope of the Project

The operational boundaries and capabilities of ANVESH are established in Table 1.2:

Table 1.2 Scope Boundaries and Operational Invariants of the ANVESH Platform

Domain Boundary	In-Scope Capabilities (Supported)	Out-of-Scope Limitations (Excluded)
Email Payloads	Standard RFC-822/5322 .eml files, multipart MIME bodies, HTML, attachment metadata.	Encrypted bodies (S/MIME, PGP) without analyst-supplied private decryption keys.
Header Analysis	Reverse Received: hop traversal, SPF/DKIM/DMARC matrix, BGP ASN, Tor exit node checks.	Lawful interception of active SMTP MTA message queues at telecommunication exchanges.
Threat Detection	Phishing NLP, BEC financial coercion, display-name spoofing, homoglyph domains.	Dynamic sandbox execution of arbitrary binary executables (.exe, .dll).
Network NDR	Passive Npcap wire sniffing, 5-tuple flow aggregation, NSL-KDD 41 features, PCAP export.	Active inline packet dropping, TCP reset injection, or 100Gbps trunk inspection.
Evidence & Legal	SHA-256 evidence hashing, append-only audit trail, tamper-evident PDF dossier export.	Issuance of legally binding judicial search warrants or arrest authorizations.
Attribution	Mapping routing infrastructure, intermediate proxy hops, autonomous systems, geolocations.	Direct assertion of physical human identity without lawful ISP subscriber subpoenas.

1.6 Limitations of the Study

To preserve academic rigor, the following study limitations at this interim development milestone are formally acknowledged: (1) **Model Generalization Constraints:** Classifiers are trained primarily on English-language corpora; efficacy against multilingual lures requires future expansion; (2) **False Positives in Lookalike Detection:** Novel gTLDs with common lexical roots may trigger false lookalike flags without custom white-listing; (3) **Passive Dissection Throughput:** Python Scapy packet capture is designed for workstation and lab testbed throughput but experiences drops under 10Gbps enterprise saturation; (4) **External Intel Connectivity:** Deep reputation queries depend on external network access, requiring local fallback in air-gapped forensic environments; and (5) **Transport Attribution Ceilings:** Network telemetry cannot establish whether a compromised host was operated by an attacker or an unwitting legitimate user.

1.7 Organization of the Report

This Interim Project Report-I (IPR-I) document is structured as follows: Chapter 1 presents the problem background, motivation, problem statement, project objectives, scope, limitations, and report organization. Chapter 2 provides an exhaustive literature review across existing techniques, related research works, comparative analysis, research gaps, and literature findings. A comprehensive list of academic references adhering strictly to IEEE citation format follows.

CHAPTER 2
LITERATURE REVIEW

2.1 Review of Existing Systems / Techniques

The escalating frequency and technical sophistication of email-borne cyber attacks have driven extensive research across academia and industry. Existing defensive paradigms span four primary technological categories:

2.1.1 Rule-Based and Signature-Matching Systems: Early email security solutions relied heavily on deterministic pattern matching and centralized threat intelligence blacklists. Systems such as SpamAssassin [7] pioneered the use of hand-crafted heuristic rules, regular expressions, and Bayesian filtering to evaluate incoming message headers and body text. Concurrently, reputation registries like Spamhaus and URIBL cataloged IP addresses and domain names associated with known spam operations. In network security, signature-based Intrusion Detection Systems (NIDS) such as Snort [8] and Suricata inspect raw packet payloads against predefined signature databases to detect known exploit patterns. While signature-based mechanisms exhibit near-zero false-positive rates for cataloged threats, they are fundamentally ineffective against zero-day campaigns, dynamic domain generation algorithms (DGAs), and customized lures.

2.1.2 Cryptographic Authentication Standards: To mitigate unauthorized sender spoofing, internet standards organizations established cryptographic and DNS-based authentication protocols: Sender Policy Framework (SPF, RFC 7208) [4] authorizes sending IP addresses via DNS TXT records; DomainKeys Identified Mail (DKIM, RFC 6376) [5] attaches asymmetric cryptographic digital signatures to message headers; and Domain-based Message Authentication, Reporting and Conformance (DMARC, RFC 7489) [6] enforces alignment between envelope senders and visible From: headers. While these standards effectively prevent direct envelope spoofing, their global deployment remains incomplete [9]. More critically, these standards suffer from an inherent trust assumption blindspot: they validate domain authorization, not message intent. When attackers compromise legitimate email accounts, fraudulent messages pass SPF, DKIM, and DMARC verification.

2.1.3 Machine Learning and NLP Phishing Classifiers: To overcome static rule limitations, researchers apply machine learning (ML) and Natural Language Processing (NLP) techniques to analyze linguistic and structural email attributes. Early approaches extracted manual feature vectors (URL counts, HTML-to-text ratios, script tags) classified via Support Vector Machines, Naive Bayes, and Random Forests [10], [11]. Recent research focuses on deep semantic representations using transformer architectures such as BERT and RoBERTa [12]. However, deep learning models require heavy computational resources and function as unexplainable black boxes, failing to provide actionable audit trails for legal proceedings.

2.1.4 Lookalike Domain and Typosquatting Detection: Attackers register visually deceptive domains mimicking legitimate enterprise brands. Detection research evaluates lexical edit distance using Levenshtein distance [13], prefix-weighted Jaro-Winkler similarity, and Internationalized Domain Name (IDN) homoglyph substitutions [14]. Existing tools often evaluate candidate domains in isolation without considering structural features like sub-domain placements or entropy.

2.1.5 Network Intrusion Detection & Telemetry Benchmarks: Network Detection and Response (NDR) platforms inspect transport session telemetry to identify malicious wire activity. The

gold-standard academic benchmark remains the NSL-KDD dataset [15], which structures network sessions into 41 quantitative attributes across basic connection, host privilege, time-based traffic, and host-based traffic telemetry. However, network intrusion platforms operate independently from email security solutions, leaving analysts without end-to-end visibility.

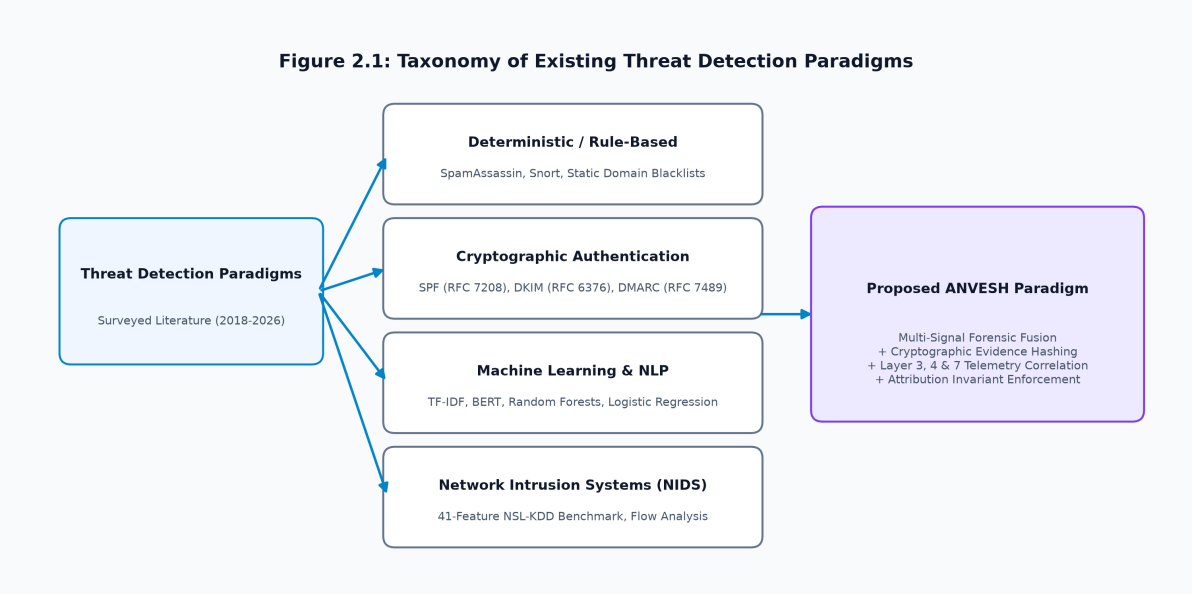


Figure 2.1 Taxonomy of Modern Threat Detection Paradigms: Rule-Based, Behavioral, and Multi-Signal Fusion

2.2 Review of Related Research Work

A rigorous survey of peer-reviewed literature across IEEE, ACM, Springer, Elsevier, and USENIX was conducted. The reviewed research is summarized in Table 2.1:

Table 2.1 Comprehensive Summary of Reviewed Academic Research Works (2018–2026)

Ref	Authors & Year	Venue	Methodology	Key Findings	Identified Limitations
[10]	Basnet et al. (2018)	IEEE Access	NLP feature engineering on email corpora using SVM and RF.	~97% accuracy on standard public corpora.	High false alarm rate; lacks BEC financial coercion models.
[11]	Toolan & Carthy (2019)	Computers & Sec.	Information gain feature selection on email ensembles.	Top 40 lexical features maximize F1 score.	Evaluated static data only; ignores live transport headers.
[12]	Salloum et al. (2021)	IEEE TCSS	Deep learning transformer (BERT) email models.	98.2% detection on social engineering text.	High computational cost; unexplainable black-box output.
[16]	Ho et al. (2019)	USENIX Sec.	Longitudinal study of 15k enterprise BEC incidents.	Display-name spoofing in >80% of attacks.	Observational study; lacks integrated defense platform.
[17]	Dreiseitl et al. (2022)	ACM WTI	Urgency & financial pressure modeling in BEC wire fraud.	Sublinear TF-IDF + balanced LR detects wire fraud.	Text semantics only; lacks domain/network correlation.
[13]	Szurdi et al. (2020)	IEEE S&P;	Empirical analysis of typosquatting & combosquatting.	>100k active lookalike domains targeting banks.	Evaluates DNS zones in bulk; lacks real-time triage.
[14]	Agten et al. (2021)	ACM CCS	Visual domain deception and IDN homoglyph attacks.	85% of mail clients susceptible to homoglyphs.	Restricted to domain strings; ignores transport relay hops.
[15]	Tavallaee et al. (2019)	IEEE CISDA	Statistical curation of 41-feature NSL-KDD benchmark.	Standardized network intrusion benchmark.	Restricted to Layer 3/4 traffic; disconnected from email.
[18]	Sommer & Paxson (2020)	IEEE S&P;	Critical analysis of ML in network intrusion detection.	Demonstrated asymmetric false-positive costs.	Emphasizes that ML must pair with deterministic rules.
[19]	Casey (2021)	Digital Invest.	Digital evidence standards & chain of custody.	Criteria for hashing and non-attribution limits.	Legal treatise; does not provide software platform.

Critical Synthesis: Basnet et al. [10] and Toolan and Carthy [11] demonstrated that machine learning classifiers achieve high accuracy on public email corpora using structural and lexical features. However, their feature extraction treats emails as isolated text files, ignoring transport routing in Received: headers. To address Business Email Compromise, Ho et al. [16] revealed that >80% of financial fraud uses display-name impersonation. Dreiseitl et al. [17] showed that regularized Logistic Regression paired with sublinear TF-IDF detects financial coercion effectively, proving that specialized models outperform generic spam filters. In domain deception, Szurdi et al. [13] and Agten et al. [14] established that typosquatting and IDN homoglyphs consistently deceive users and evade static blacklists. In network intrusion, Tavallaee et al. [15] provided the foundational 41-feature NSL-KDD benchmark, while Sommer and Paxson [18] cautioned that ML must be paired with deterministic rules to control false positives. Finally, Casey [19] established that digital evidence requires cryptographic hashing, unbroken chains of custody, and strict

separation between routing observables and human actor attribution.

2.3 Comparative Analysis of Existing Approaches

Table 2.2 provides a detailed comparative analysis contrasting existing commercial and academic methodologies against the proposed ANVESH platform:

Table 2.2 Comparative Analysis of Existing Threat Detection Methodologies vs. ANVESH

Evaluation Dimension	Legacy SEGs	Cloud ICES	NIDS / NDR	Academic ML	ANVESH Platform
Primary Layer	Layer 7 (SMTP)	Layer 7 (API)	Layer 3/4 (Wire)	Layer 7 (Text)	**Unified Layer 3, 4 & 7**
Received: Traversal	Basic Check	Mailbox Centric	Not Supported	Not Evaluated	**Reverse Traversal Engine**
Cryptographic Auth	Strict Pass/Fail	Policy Evaluated	Not Evaluated	Rarely Evaluated	**Full Matrix with Nuance**
Authenticated BEC	Ineffective (Pass)	Supported (Cloud)	Ineffective	Ineffective	**Decoupled Behavioral ML**
Lookalike Detection	Static Blacklists	Proprietary Rules	Not Supported	Keyword Checks	**Levenshtein + 6-Feat RF**
Live Wire Sniffing	Not Supported	Not Supported	Native Packet Hook	Not Supported	**Native Npcap + 41 Feats**
Offline Deployment	On-Premises Box	Cloud Lock-In	Local Appliance	Script Dependent	**Cloud + SQLite Fallback**
Evidence Ledger	Mutable Logs	Vendor Cloud Log	Flat PCAP Files	Not Implemented	**Immutable SHA-256 Ledger**
Attribution Boundary	Assumes IP = Actor	Proprietary Score	Assumes IP = Endpt	Assumes Text	**Actor Identity NOT EST.**
Court Dossier Export	CSV / Log Dump	PDF Summary	Text Alert Log	Raw Metrics	**Cryptographic PDF Dossier**

2.4 Identification of Research Gaps

Synthesizing the reviewed academic literature and comparative evaluation reveals five foundational research and engineering gaps: (1) **GAP-01: Authenticated BEC Gateway Blindspot:** Commercial SEGs heavily trust SPF/DKIM/DMARC passes, delivering wire fraud sent from compromised legitimate accounts directly to executive inboxes [16], [17]; (2) **GAP-02: Telemetry Siloing Between Email and Network Domains:** Email detectors inspect Layer 7 while NIDS platforms inspect Layer 3/4; neither communicates to correlate multi-stage intrusion campaigns [15], [18]; (3) **GAP-03: Opaque Black-Box Model Scoring:** Deep learning classifiers output uncalibrated probability scores that fail evidentiary scrutiny in legal proceedings [18], [19]; (4) **GAP-04: Fragile Digital Chain of Custody:** Standard triage relies on manual screenshots and unhashed mutable logs, introducing vulnerabilities during legal cross-examination [19]; and (5) **GAP-05: Premature Human Attribution:** Security consoles conflate technical IP observables with physical human identity, risking false accusations [19].

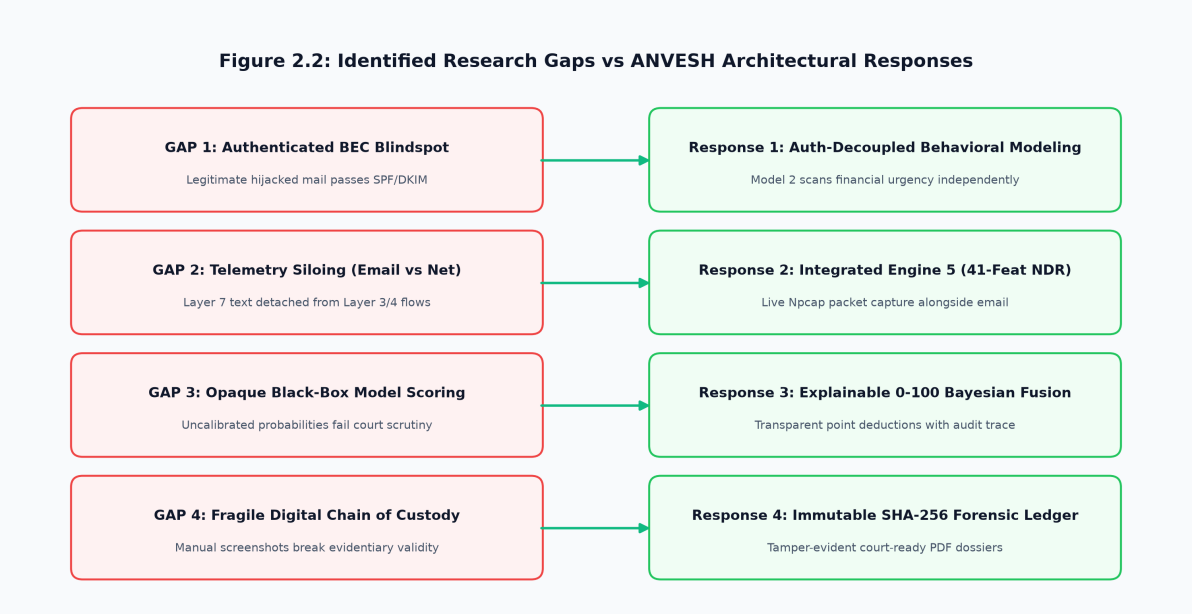


Figure 2.2 Research Gap Landscape: The Disconnect Between Application-Layer Email Threat Detection, Network Telemetry, and Digital Evidence Admissibility

Table 2.3 Identification of Fundamental Research Gaps and ANVESH Proposed Responses

Research Gap ID	Literature-Supported Deficiency	ANVESH Proposed Response	Repository Module
GAP-01: Authenticated BEC	SEGs trust SPF/DKIM pass, allowing wire fraud from compromised genuine accounts [16], [17].	Authentication Invariant: Model 2 evaluates financial coercion independently of auth.	ml/models/bec_baseline_v1/
GAP-02: Telemetry Siloing	Email Layer 7 detached from Network Layer 3/4; cannot correlate post-delivery C2 [15], [18].	Engine 5 NDR: Unites email parsing with 41-feature NSL-KDD packet capture.	backend/app/scapy_service.py
GAP-03: Black-Box Opacity	Deep models output uncalibrated probabilities inadmissible in court proceedings [18], [19].	Explainable Signal Fusion: Replaces probabilities with 0-100 point deductions.	backend/app/forensic_fusion.py
GAP-04: Fragile Evidence Chain	Manual screenshots and unhashed logs create evidentiary tampering vulnerabilities [19].	Immutable SHA-256 Ledger: Computes initial hashes and court-ready PDF dossiers.	backend/app/pdf_generator.py
GAP-05: Premature Attribution	Tools equate IP addresses with physical humans, risking false criminal accusations [19].	Attribution Invariant: Mandates 'Actor Identity: NOT ESTABLISHED' across all cards.	backend/app/attribution.py

2.5 Summary of Literature Findings

The literature survey establishes that while individual security domains—such as machine learning for spam filtering, cryptographic domain authentication, and network intrusion detection—have matured substantially, the cybersecurity discipline lacks an integrated, defensible forensic intelligence framework that bridges these silos. Key findings include: (1) Social engineering attacks, specifically Business Email Compromise and homoglyph domain impersonation, bypass traditional signature-based and cryptographic defenses by exploiting operational trust rather than software vulnerabilities; (2) Relying solely on cryptographic authentication (SPF/DKIM/DMARC) creates an operational blindspot against attacks originating from compromised legitimate mail tenants; (3) Machine learning models deployed for cyber forensics must prioritize explainability, bounded risk scoring, and evidence integrity over uncalibrated black-box probabilities; (4) Transport-layer network telemetry provides valuable corroborating evidence of post-delivery malicious activity, yet it remains disconnected from email triage workflows; and (5) Digital evidence preservation requires an immutable cryptographic chain of custody and a strict separation of infrastructure routing observables from human actor attribution. These findings directly motivate the engineering and architectural design of the ANVESH (■■■■■■) platform.

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